Solution

Task1

Sort the array [2, 0, 1, 2, 0, 1] using a while loop so that all 0s come first, then 1s, then 2s Solution:-

```
arr = [2, 0, 1, 2, 0, 1]
counts = [0, 0, 0]
for num in arr:
  if num < len(counts):
    counts[num] += 1
k = 0
for j in range(len(counts)):
  while counts[j] > 0:
    arr[k] = j
    k += 1
    counts[j] -= 1
for element in arr:
  print(element)
```

output:-

```
...   0
      0
      1
      1
      2
      2
```

Task 2

Generate and print 10 random numbers between 1 and 100 using a for loop Solution:-

```
import random
print("Random Numbers:")
for i in range(10):
 print(random.randint(1, 100))
```

```
Generating 10 random numbers from 1 to 100:
44
27
9
84
70
30
67
35
37
81
```

Task 3

Industry Example + Four Levels of Intelligence

```
Industry: Healthcare

1. Statistics:
- Used to analyze patient data such as average recovery time, mortality rates, and disease trends.

2. Artificial Intelligence (AI):
- Rule-based systems for medical diagnosis (e.g., if symptoms A and B → disease X).

3. Machine Learning (ML):
- Models predict diseases like diabetes or heart disease using patient data.

4. Deep Learning (DL):
- CNNs analyze X-rays, MRIs, and CT scans for cancer and brain tumors.
```

Task 4

Data Preprocessing on Titanic Dataset

Steps Covered

- Load dataset

import pandas as pd

import numpy as np

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"

df = pd.read_csv(url)

df.head()

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

- **Handle missing values**

df.isnull().sum()

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

- **Numerical & categorical imputation**

```
# Fill numerical columns with mean
df['Age'].fillna(df['Age'].mean(), inplace=True)
df['Fare'].fillna(df['Fare'].mean(), inplace=True)

# Fill categorical columns with mode
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

	PassengerId	Survived	Pclass	Name	Age	SibSp	Parch	Ticket	Fare	Cabin	Sex_male	Em
0	1	0	3	Braund, Mr. Owen Harris	22.0	1	0	A/5 21171	7.2500	NaN	True	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	0	PC 17599	71.2833	C85	False	
2	3	1	3	Heikkinen, Miss. Laina	26.0	0	0	STON/O2. 3101282	7.9250	NaN	False	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0	113803	53.1000	C123	False	
4	5	0	3	Allen, Mr. William Henry	35.0	0	0	373450	8.0500	NaN	True	

- **Encode categorical variables**

Convert categorical variables into numbers

df = pd.get_dummies(df, columns=['Sex', 'Embarked'], drop_first=True)

df.drop(['Name', 'Ticket', 'Cabin'], axis=1, inplace=True)

```
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value

  df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	Sex_male	Embarked_Q	Embarked_S
0	1	0	3	22.0	1	0	7.2500	True	False	True
1	2	1	1	38.0	1	0	71.2833	False	False	False
2	3	1	3	26.0	0	0	7.9250	False	False	True
3	4	1	1	35.0	1	0	53.1000	False	False	True
4	5	0	3	35.0	0	0	8.0500	True	False	True

- **Final ML-ready dataset**

df.head()

print("Final Dataset Shape:", df.shape)

Final Dataset Shape: (891, 9)

	PassengerId	Survived	Pclass	Name	Age	SibSp	Parch	Ticket	Fare	Cabin	Sex_male	Em
0	1	0	3	Braund, Mr. Owen Harris	22.0	1	0	A/5 21171	7.2500	NaN	True	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	0	PC 17599	71.2833	C85	False	
2	3	1	3	Heikkinen, Miss. Laina	26.0	0	0	STON/O2. 3101282	7.9250	NaN	False	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0	113803	53.1000	C123	False	
4	5	0	3	Allen, Mr. William Henry	35.0	0	0	373450	8.0500	NaN	True	